COMPACT AI HACKATHON

Project Portfolio

On-Device Crop Disease Detection for Indian Farmers

A compact, offline AI model that helps a farmer diagnose a diseased crop from a single photo — no internet required. A working prototype is already built (Section 6).

1. Team Details

Name	Role	Key Responsibility
[Member 1]	Team Lead	Coordination, final submission, presentation
[Member 2]	Research & Content	Problem research, impact data, writing
[Member 3]	ML / Prototype	Model training, ONNX export, testing
[Member 4]	App / Front-end	App UI and camera flow (or demo build)
[Member 5]	Design & QA	Portfolio design, review, demo video

Contact: [team email] • College/Institute: [name] • Submission date: 15 October 2026

2. Problem Statement

Every season, Indian farmers lose a large share of their crop to pests and diseases. Most of them have no quick, affordable way to find out what is wrong with their plants. Diagnosis today depends on agricultural extension officers who are few and far between, or on guesswork — which often leads to the wrong pesticide, over-spraying, higher cost and lower income. A farmer in a remote village may own a smartphone but have weak or no internet, so cloud-based AI tools are of little use.

This project addresses that gap with a compact, offline AI model that runs entirely on the farmer's phone.

3. Why It Matters

The scale of the loss is well documented in Indian agricultural research:

- Pre-harvest crop loss due to pests and diseases in India is estimated at around 20%.
- Crop loss due to insect pests is estimated at about 25% in rice and maize, 15% in pulses, and up to 50% in cotton.
- Plant-parasitic nematodes alone cause 21.3% crop losses, worth roughly ₹10,200 crore (about US$1.58 billion) every year.
- In a survey of Indian tomato farmers, the average yield loss from pests and diseases was about 11 tonnes per hectare.
- The large majority of Indian farmers are small and marginal landholders (around 86%, Agriculture Census 2015–16), who can least afford a failed season.

A fast, offline diagnosis directly reduces wrong pesticide use, cuts input cost, and protects yield — with no dependence on connectivity or an officer's visit.

4. Proposed Solution

A lightweight application that lets a farmer photograph a diseased leaf or plant and, within a second and completely offline, receive:

- the likely disease name,
- a confidence score,
- simple, farmer-friendly treatment advice (starting in Hindi and English).

No account, no upload, no internet — the photo is analysed on the device itself.

5. The Compact AI Angle

The heart of the solution is a small model that runs on-device. This is what makes it a compact-AI solution rather than a cloud service:

- Model size of about 4.4 MB, so it fits comfortably on low-end phones.
- Inference in about 1.5 milliseconds per image on a CPU, with no network round-trip.
- Works fully offline — designed for rural areas with weak connectivity.
- No farmer data ever leaves the device, which is good for privacy and trust.

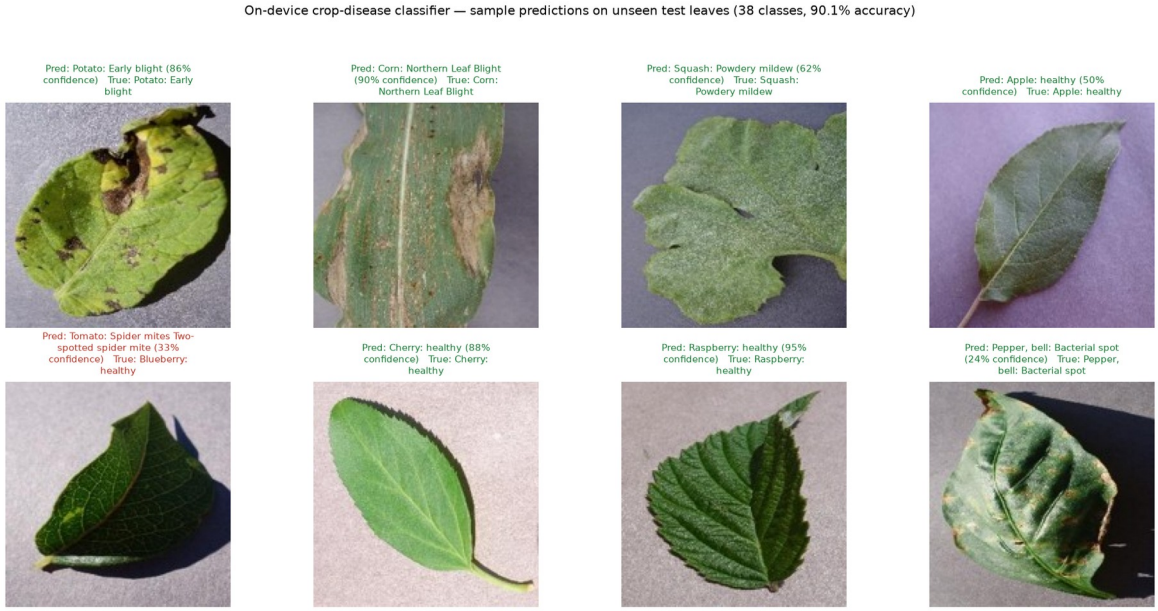
Efficiency here is not an afterthought; it is the design constraint that makes the tool usable where it is needed most.

6. Results — Prototype Already Built

We did not stop at a plan. A working model has been trained, exported and tested. It classifies 38 crop-and-disease combinations at 90.1% accuracy on 1,140 unseen test leaves.

Metric	Measured value
Classes (crop + disease)	38
Validation accuracy	90.1% on 1,140 held-out images
Model size (ONNX)	4.36 MB
Inference latency (CPU)	~1.5 ms per image
Runtime	ONNX Runtime — runs on-device / edge

Sample predictions on unseen test leaves (green = correct, red = wrong):



7. How It Works (Technical Approach)

Training a deep network from scratch needs a GPU. To keep the prototype practical on ordinary hardware, we used transfer learning with a frozen backbone:

- A MobileNetV3-Small network, pre-trained on ImageNet, is used as a fixed feature extractor.
- Each leaf image is passed through it once to get a 576-dimensional feature vector (forward pass only).
- A small classifier head is trained on those cached features (a 'linear probe') — this trains in seconds.
- The head is folded back onto the backbone and the whole model is exported to ONNX for on-device use.

8. Tech Stack

Layer	Choice
Model	MobileNetV3-Small (ImageNet) as a frozen feature extractor + trained classifier head
On-device runtime	ONNX Runtime (CPU) for offline inference
App	Android (Kotlin) or Flutter camera app; Gradio web demo for the prototype
Advice layer	Small offline rules / lookup for treatment suggestions (Hindi + English)
Training & testing	PyTorch feature extraction + head training; ONNX export; 5,300 training images

9. MVP Scope

Done

- A trained 38-class classifier at 90.1% accuracy, exported to a 4.4 MB ONNX model.
- An offline inference script and a web demo that returns the top-3 diseases with confidence.

Next

- Wrap the model in an Android app with camera capture.
- Add treatment advice in Hindi and regional languages.
- Static int8 quantization with calibration to shrink the model further.
- Test on real field photos, not only lab images.

10. Impact & Beneficiaries

- Small and marginal farmers, who form the majority of Indian farm holdings.
- Reduced pesticide spend and safer produce through correct, early diagnosis.
- Faster help where extension officers are unavailable or far away.
- Works in low-connectivity rural areas where cloud tools fail.
- Scalable later to more crops, more languages, and advisory integration.

11. Execution Roadmap & Next Steps

Days	Milestone
Day 1–2	Research, choose crop + disease list, pick a model approach
Day 3–4	Collect data, build and train the compact model
Day 5–6	Export to ONNX, measure size / accuracy / latency
Day 7–9	Build the demo, write and design the portfolio
Day 10–12	Team review, feedback, corrections
Day 13	Final proofread and online submission

12. Risks & Mitigation

- Dataset bias — PlantVillage images are lab-quality; real field photos differ. Mitigation: add field images and state the limitation honestly.
- int8 quantization lowered accuracy in our test — use static quantization with a calibration set next.
- Team is beginner-level in coding — we used a frozen pretrained backbone and a no-code web demo rather than training from scratch.
- Time is short — keep the first version to a single, well-tested model.

13. Sources

Crop-loss figures cited in Section 3 are drawn from published Indian agricultural research, including:

- Pre- and Post-Harvest Losses study (pest and disease loss estimates; Dhaliwal et al., 2010).
- National Academy Science Letters (2020) — crop loss estimation due to plant-parasitic nematodes in India.
- Tomato pest and disease management study for India and Bangladesh (2023).
- Agriculture Census of India (2015–16) — small and marginal holdings share.

Model data: PlantVillage dataset — Mohanty, Hughes & Salathé (2016), Frontiers in Plant Science.